

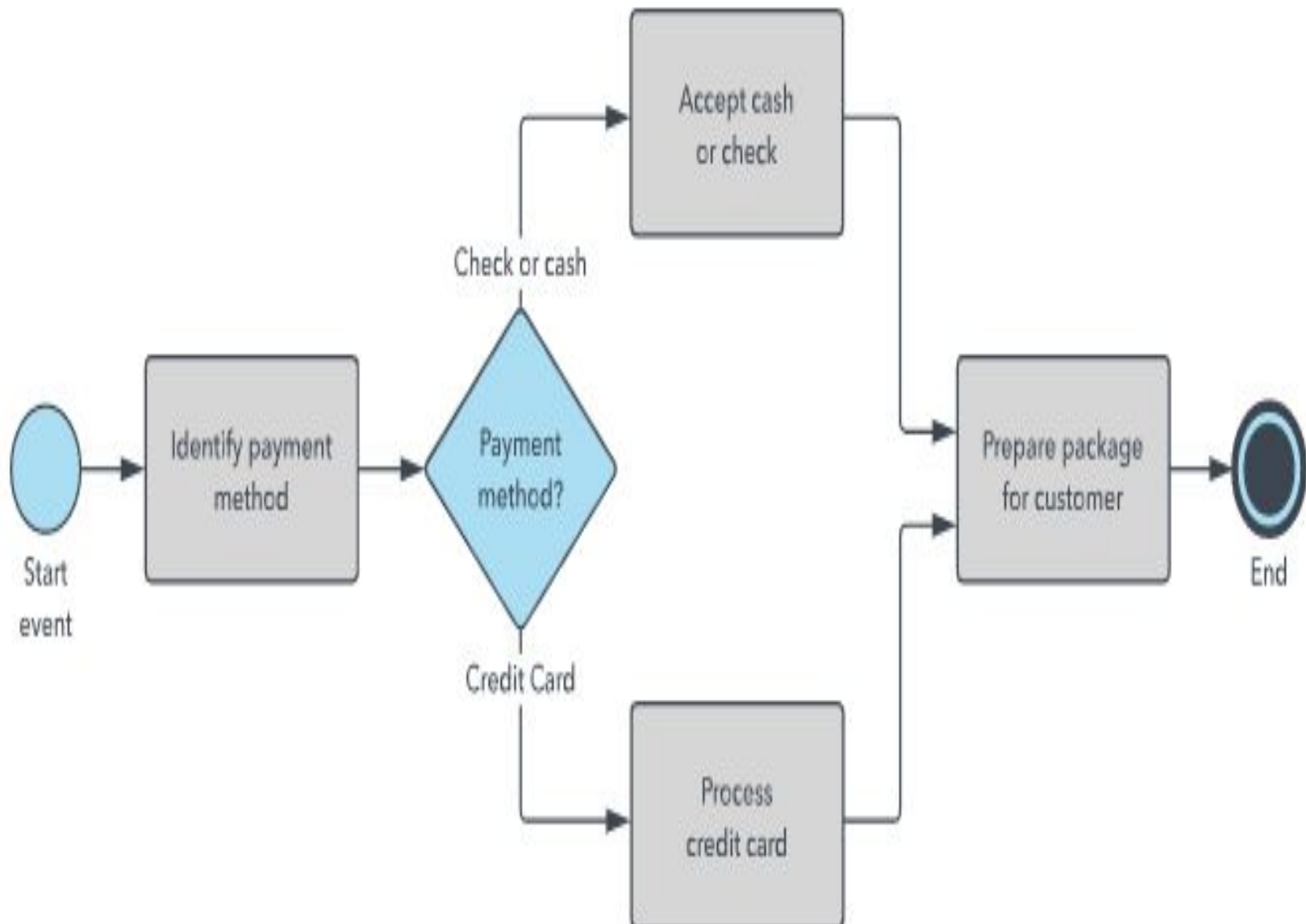
Modeling Business processes (Part 1)- continuation...

- **Definition, purpose and benefits of BPMN**
- **Stages in development & creating a business process model**
- **Symbols and notation**

What is BPMN?

Business Process Modeling Notation (BPMN) is a flowchart technique for modeling the steps of a planned business process from start to finish. It is an important part of Business Process Management because it visually illustrates the sequence of business activities and information flows required to complete a process.





Purpose and benefits

- ✓ *to increase efficiency*
- ✓ *Adapt to changing circumstances or obtain a competitive advantage*
- ✓ *to obtain understanding of the steps through an easy-to-understand visual representation*
- ✓ *At a deeper level, it's aimed at the people who will carry out the process, providing enough detail to ensure flawless execution.*
- ✓ *The diagramming may be much simpler to comprehend.*

- ✓ *It establishes a consistent language for all stakeholders, technical and non-technical alike: business analysts, process participants, managers and technical developers, as well as external teams and consultants.*
- ✓ *It fills in the gaps between process intent and execution by providing enough depth and clarity into the sequence of business processes.*
- ✓ *Enables easier communication and collaboration in order to achieve the goal of an efficient procedure that yields a high-quality outcome*

Stages in Development of BPMN:

- Identify the process and produce an 'as is' or baseline model.
- Review, analyze and update the 'as is' process model.
- Design the 'to be' model.
- Test and implement the 'to be'.
- Continuously update and improve the new model.

Defining the As Is Process

The As Is process shows your current processes - what your organization currently does.

It is important to bear in mind that this As Is analysis may only show you what can be improved, but not necessarily how. Mapping the As Is process will only reveal your business processes as they are today.

Defining the To Be Process

The *To Be* process shows your proposed future processes - what your organization plans to do.

By creating a process model of the To Be you can discuss it with those people who will be impacted by the changes planned and both take on board their view of the outcome of the changes and work to get their buy-in.

You can also quantify the return on investment from the changes proposed.



Creating a Business Process Model

You will need the following information before you start to construct your model:

- **The processes' desired outcome.**

- The beginning and ending points (customer need and customer need fulfillment).
- The tasks that are carried out.
- The personnel who carry out the activities.
- The documents and forms that are used and exchanged across functions, as well as from customers and suppliers.

First Draft

Because the first draft of the model will require a lot of repositioning of events and activities, choose a strategy that is adaptable and easy to adjust. Use a whiteboard and a rubber or a flipchart, pens, and sticky notes. Everyone needs to be able to see it if you're working with a group of people.

Second Draft

You can design a flowchart on generic software or specialized proprietary software once you've agreed on a sequence of activities.

You'll need to test your model with users at this point by performing 'live' observations of the sequence in action. People in focus groups or meetings almost often forget what they did or state what should happen instead of what actually happens.

Common BPMN symbols & Notations

A BPMN diagram, or business process model and notation diagram, is used to create easy-to-read business process model flowcharts that can be shared across enterprises and industries. Flow objects, linking objects, swim lanes, and artifacts are the four primary classes of BPMN diagram symbols. Below is a detailed explanation of BPMN diagram shapes and their implications.

BPMN event types

Events represent an event in a business process.

	Start event symbol - Signals the first step of a process.
	Intermediate event symbol - Represents any event that occurs between a start and end event.
	End event symbol - Signals the final step in a process.

BPMN event symbols

	Message symbol - Triggers the process, facilitates intermediate processes, or finishes the process.
	Timer symbol - A time, date, or recurring time and date triggers the process, aids intermediate processes, or completes the process.
	Escalation symbol - A step reacts on an escalation and flows to another role in the organization. This event is only used within an event sub-process. An escalation occurs when someone at a higher level of responsibility within the organization becomes involved in a process.
	Conditional symbol - A process begins or continues when a business condition or business rule is met.
	Link symbol - A sub-process that is part of a larger process.
	Error symbol - A caught error at the start, middle, or end of a process. An event sub-process with an error trigger will always interrupt its containing process.
	Cancel symbol - Reacts on a transaction that was cancelled within a sub-process. In an end event, the cancel symbol represents the triggered cancellation of a process.
	Compensation symbol - A refund that's triggered when operations partially fail.
	Signal symbol - A signal that communicates across different processes. A signal symbol can begin a process, facilitate it, or complete it.

	Multiple symbols - Multiple triggers initiating a process.
	Parallel multiple symbol - A process instance that doesn't start, continue, or end until all possible events have occurred.
	Terminate symbol - Triggers the immediate termination of a process step. All related instances are terminated at the same time.

BPMN activity symbols

Activities describe the kind of work being done in a particular process instance. There are four BPMN activity types: tasks, sub-processes, transactions, and call activities.

	Task symbol - The most basic level of an activity and cannot be broken down further. For example, a morning routine process might involve the task of turning on your computer.
	Sub-process symbol - Is a group of tasks that fit together particularly well. There are two different views of the sub-process. One is the collapsed view, which has an expandable plus sign to show more details. The other view is an expanded sub-process view, which is large enough to house all the tasks that fully describe the sub-process.
	Transaction symbol - Is a specialized sub-process that involves payment.
	Call symbol - Is a global sub-process that is reused at various points in the business flow.

BPMN gateway symbols

	Exclusive symbol - Evaluates the state of the business process and, based on the condition, breaks the flow into one or more mutually exclusive paths. For example, one report will be written if a supervisor grants approval; no report will be generated if the supervisor withholds approval.
	Event-based symbol - An event-based gateway is similar to an exclusive gateway—both involve one path in the flow. In the case of an event-based gateway, however, you are evaluating which event has occurred, not which condition is being met. For example, you might wait to send an email until the CEO has arrived at the office. If the CEO doesn't arrive, the email will remain
	unsent.
	Parallel symbol - Differs from other gateways because it's not dependent on conditions or events. Instead, parallel gateways are used to represent two concurrent tasks in a business flow. An example is a marketing department generating new sales leads and contacting existing leads at the same time.
	Inclusive symbol - Breaks the process flow into one or more flows. For example, an inclusive gateway could involve business actions taken based on survey results. One process may be triggered if the consumer is satisfied with Product A. Another flow is triggered when the consumer indicates that they are satisfied with product B, and a third process is triggered if they aren't satisfied with A.



Exclusive event-based symbol - Starts a new process instance with each occurrence of a subsequent event.



Complex symbol - These gateways are only used for the most complex flows in a business process. An ideal use case for the complex gateway is when you need multiple gateways to describe the business flow.



Parallel event-based symbol - As the name suggests, this gateway is similar to a parallel gateway. It allows for multiple processes to happen at the same time, but unlike the parallel gateway, processes are event-dependent.

3 types of Connecting objects in a BPMN diagram

	Sequence flow symbol - Connects flow objects in proper sequential order.
	Message flow symbol - Represents messages from one process participant to another.
	Association symbol - Shows relationships between artifacts and flow objects.

Swimlanes in a BPMN diagram

In a BPMN diagram, swimlanes are used to arrange different components of a process. Swimlanes divide items into lanes visually, with each step of the process assigned to its own lane. These components can be placed vertically or horizontally. Swimlanes can expose delays, inefficiencies, and the people responsible for each step in a process, in addition to organizing operations into discrete categories.



Artifacts in a BPMN diagram

Artifacts represent information that is useful to the model but not to specific process elements. Annotations, groups, and data objects are the three sorts of artifacts that can be utilized in a BPMN diagram. Each of these elements is used to supplement and characterize a BPMN process.

Annotations allow the modeller to describe more model or notation flow elements

Groups task or procedures that are important to the entire process should be organized.



Data objects represent data that is fed into the process, data that is generated by the process, data that needs to be collected, and data that needs to be saved.

	Data input symbol - Represents data requirements that tasks in the business process depend on.
	Data output symbol - Demonstrates information produced as the result of a business process.
	Data collection symbol - Signifies information collected within a business process.
	Data storage symbol - Represents the ability to store or access data that's associated with a business process.



The End

Movie ending screen background